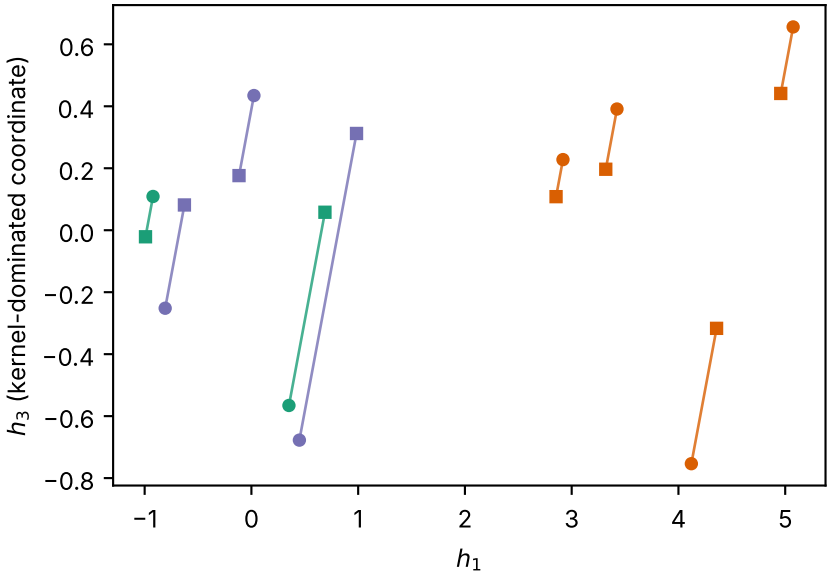
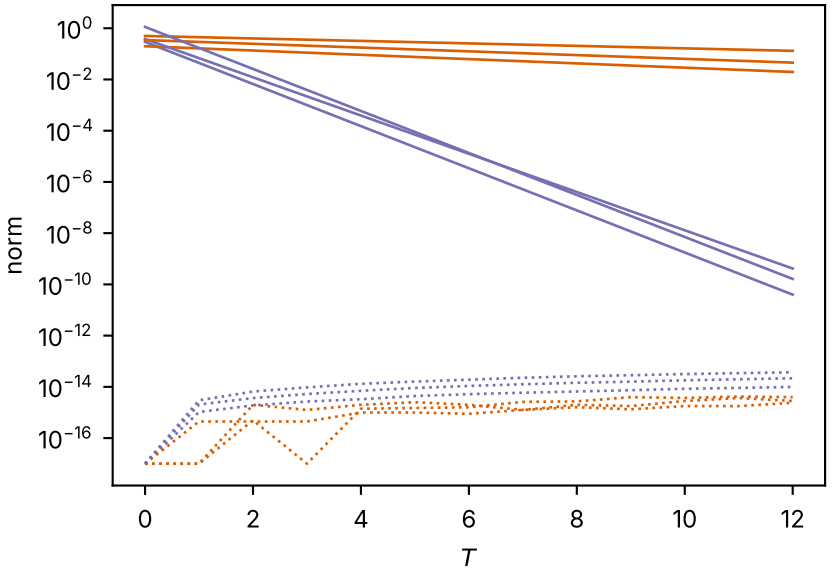

Gated recursion $h_{t+1} = (1-a)h_t + aPh_t$, $T = 0 \to 6$
(dots: h_0, squares: h_6): motion along the kernel only
h_3 (kernel-dominated coordinate)
h_1
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1
0
1
2
3
4
5


Residual $\|(I-P)h_T\|$ decays as $(1-a)^T$ (solid);
code change $\|W^\top h_T - W^\top h_0\|$ stays at 10^{-16} (dotted)
norm
T
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
0
2
4
6
8
10
12